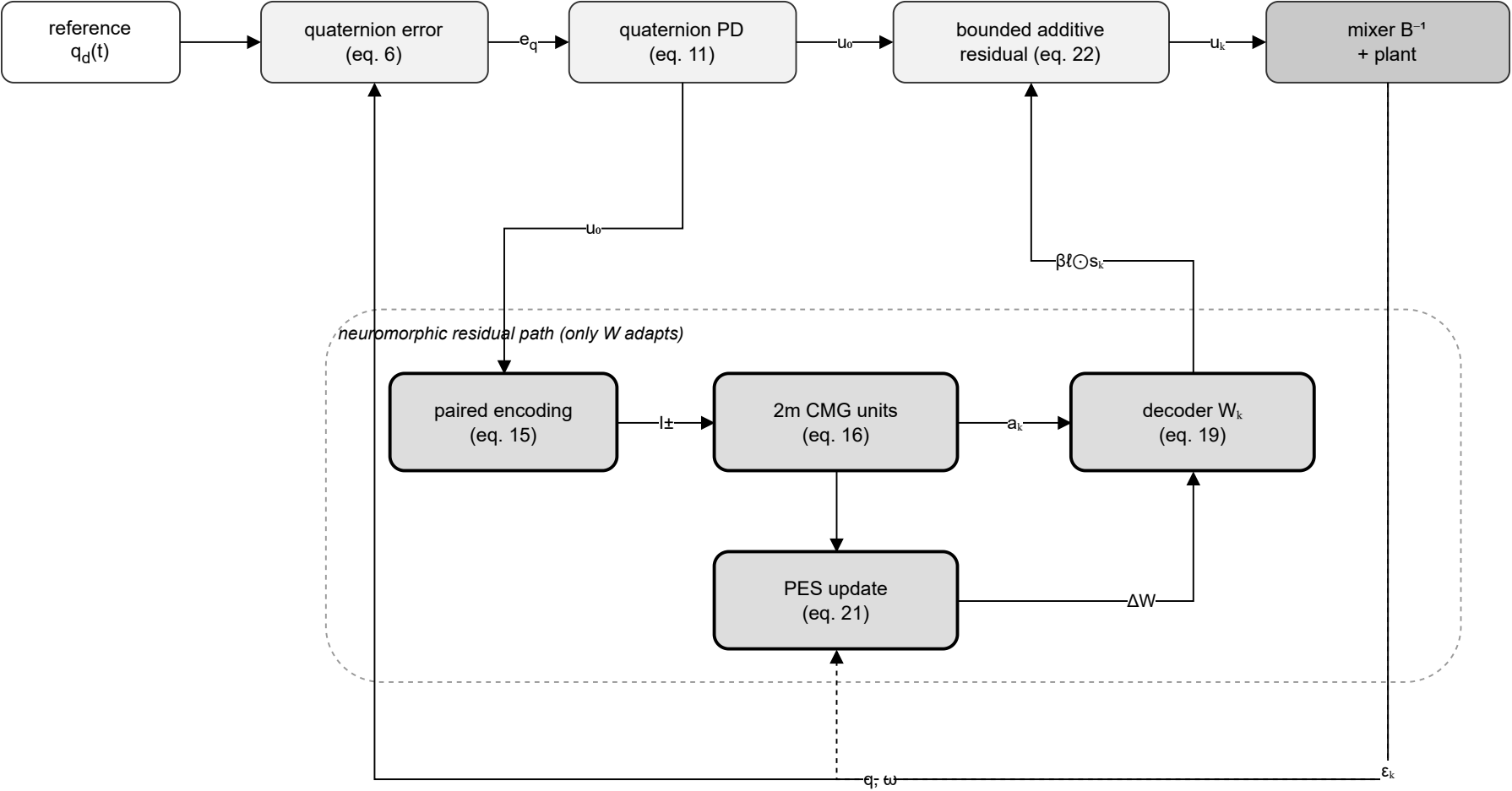
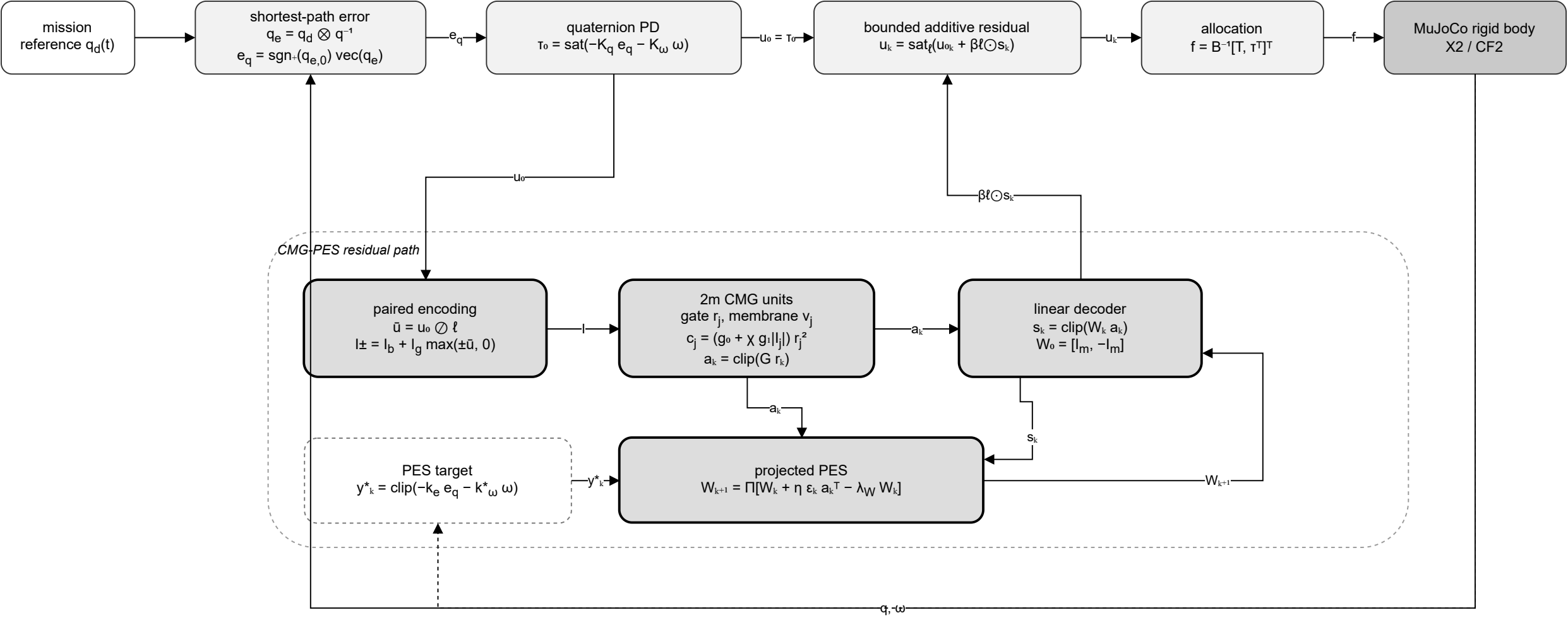
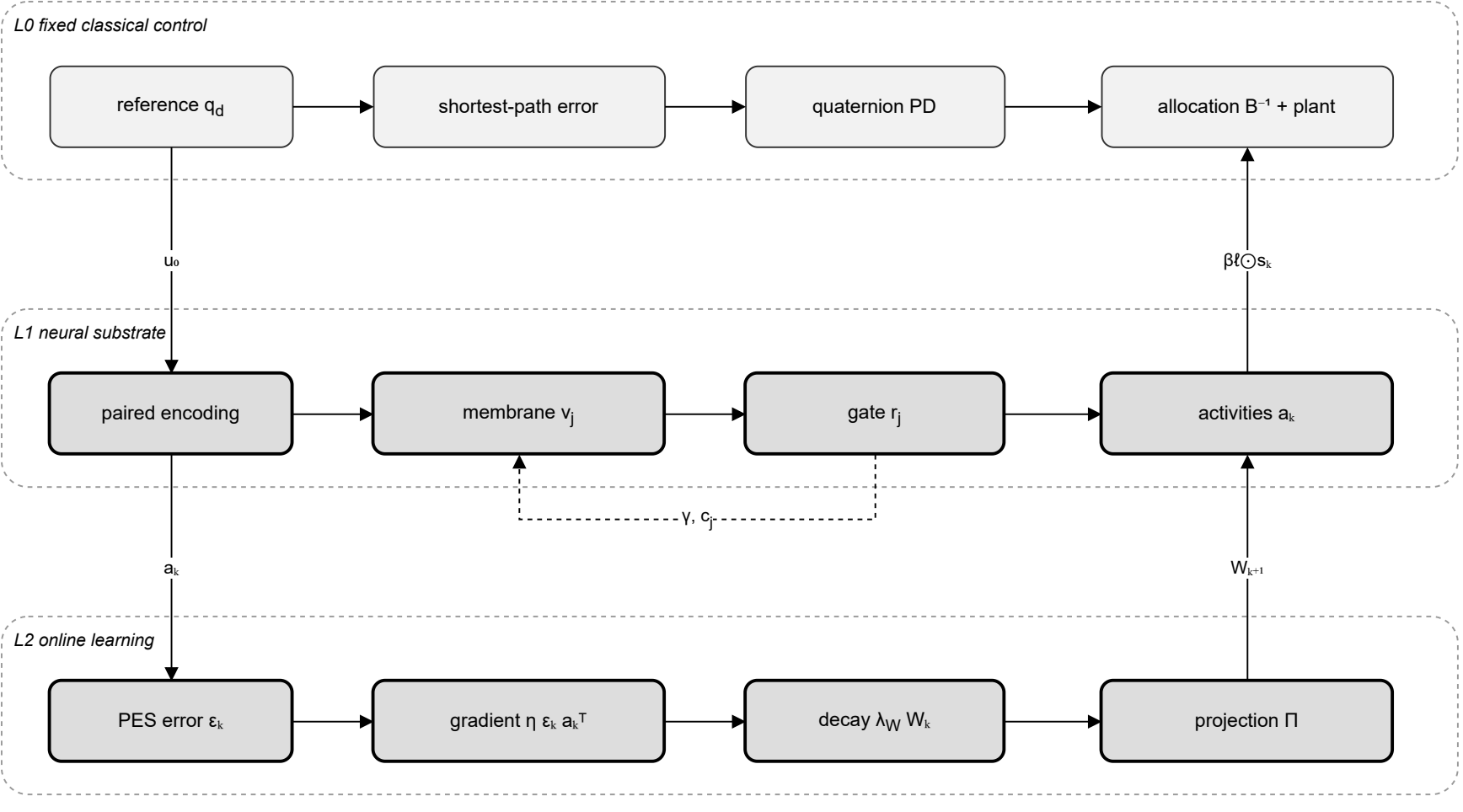


$u = \mathrm{sat}_\ell(u_0 + \beta\ell\odot s)$: the residual is ADDED to a saturated baseline, never blended with it, so β=0 recovers the baseline exactly.


mission reference $q_d(t)$
shortest-path error $q_e = q_d \otimes q^{-1}$ $e_q = \mathrm{sgn}_+(q_{e,0})\,\mathrm{vec}(q_e)$
e_q
quaternion PD $\tau_0 = \mathrm{sat}(-K_q\, e_q - K_\omega\, \omega)$
$u_0 = \tau_0$
bounded additive residual $u_k = \mathrm{sat}_\ell(u_{0k} + \beta\ell \odot s_k)$
u_k
allocation $f = B^{-1}[T, \tau^T]^T$
f
MuJoCo rigid body X2 / CF2
u_0
$\beta\ell \odot s_k$
CMG-PES residual path
paired encoding $\bar{u} = u_0 \oslash \ell$ $I_\pm = I_b + I_g \max(\pm\bar{u}, 0)$
I
2m CMG units gate r_j, membrane v_j $c_j = (g_0 + \chi\, g_I |I_j|)\, r_j^2$ $a_k = \mathrm{clip}(G\, r_k)$
a_k
linear decoder $s_k = \mathrm{clip}(W_k\, a_k)$ $W_0 = [I_m, -I_m]$
a_k
s_k
PES target $y^*_k = \mathrm{clip}(-k_e\, e_q - k^*_\omega\, \omega)$
y^*_k
projected PES $W_{k+1} = \Pi[W_k + \eta\, \varepsilon_k\, a_k^T - \lambda_W\, W_k]$
W_{k+1}
q, ω

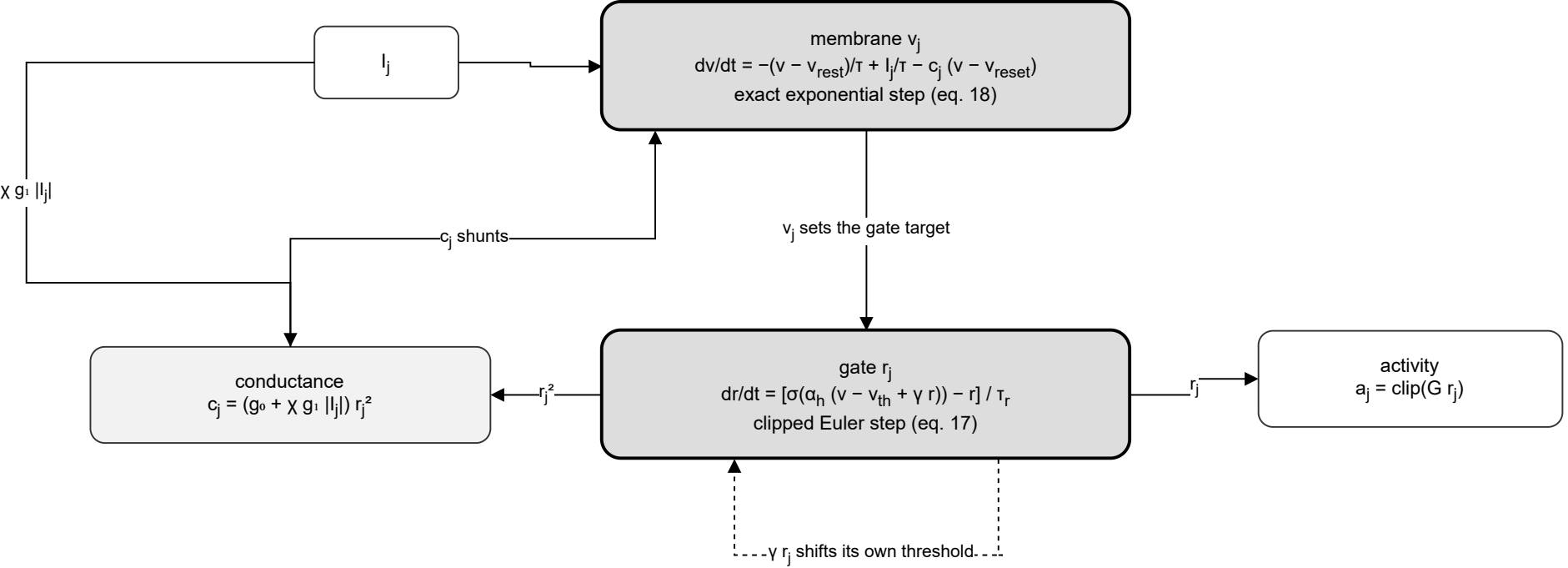


L0 fixed. Gains never adapt; the residual is added, not blended.

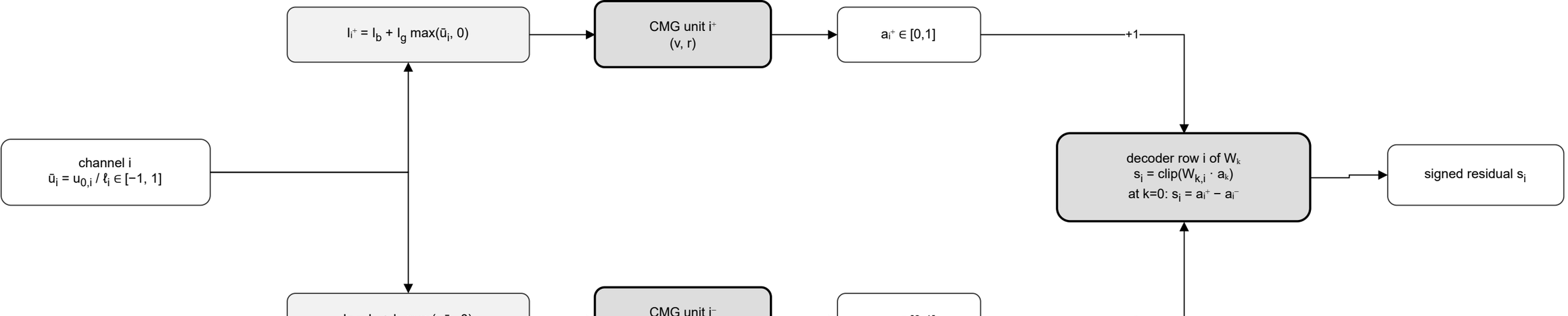
L1 substrate. Ablation knob $\chi \in \{0,1\}$ (conductance modulation).

L2 learning. Ablation knobs η, λ_W (online decoder adaptation).

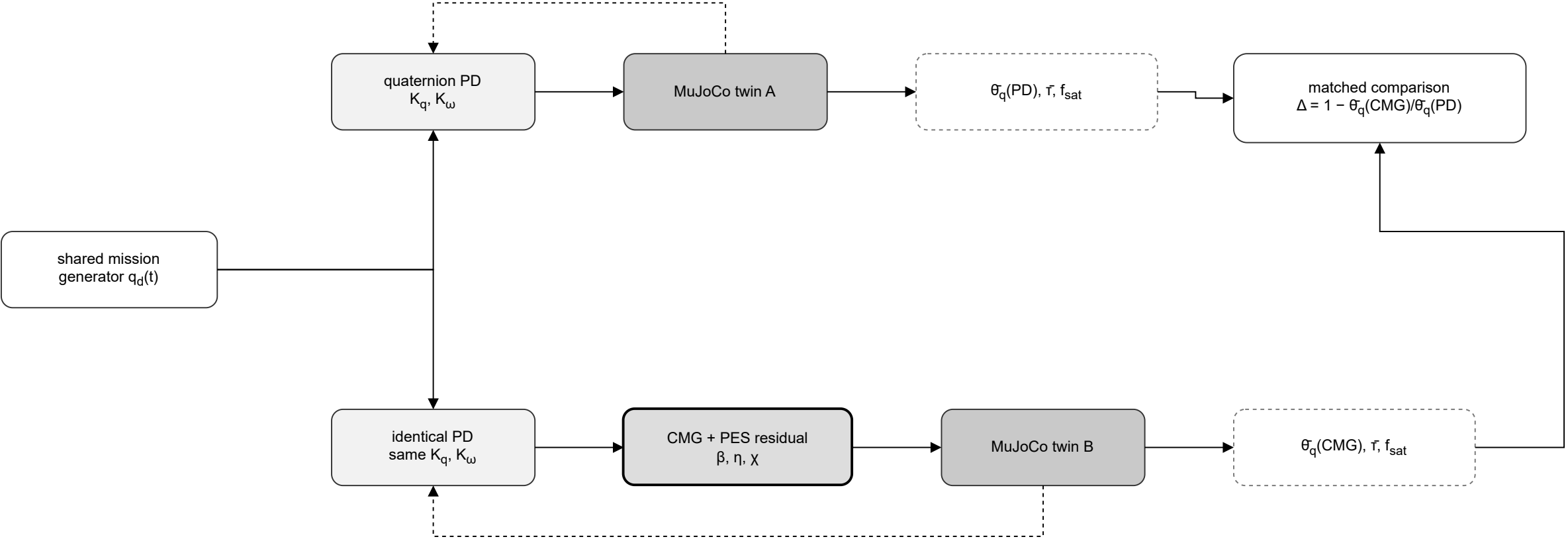
Interface knob $\beta \in [0,1]$ scales L1→L0 authority. $\beta=0$ recovers the baseline exactly; $\eta=0$ freezes $W = W_0$; $\chi=0$ removes input-dependent conductance. The three can be ablated independently.



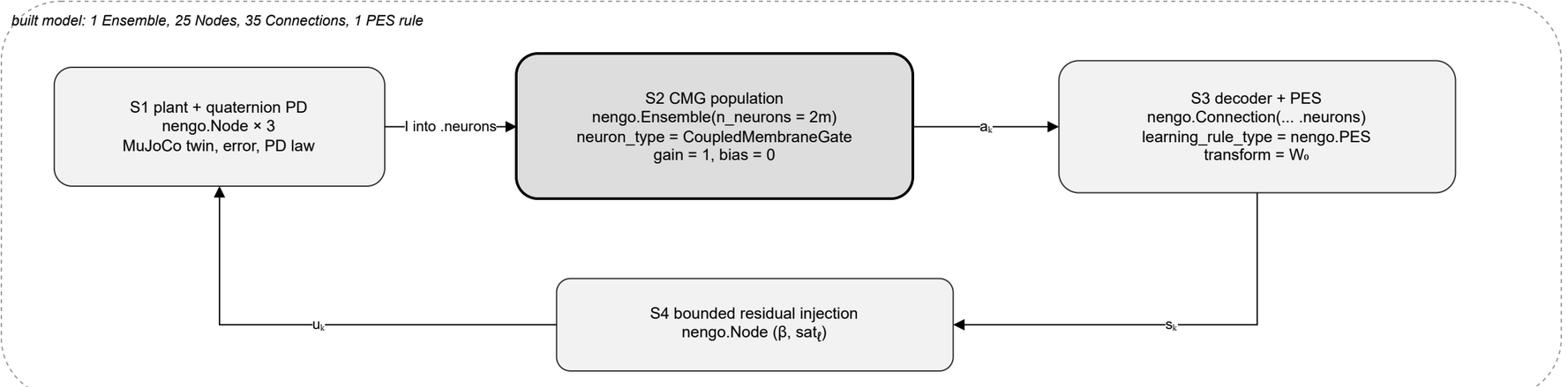
Two coupling paths make the unit coupled rather than a cascade: the gate shifts its own activation threshold (γ) and shunts the membrane through $c_j \propto r_j^2$. Continuous dynamics — no spike, no reset event — and only a_j leaves the unit. $\chi = 0$ gives the ablated variant.



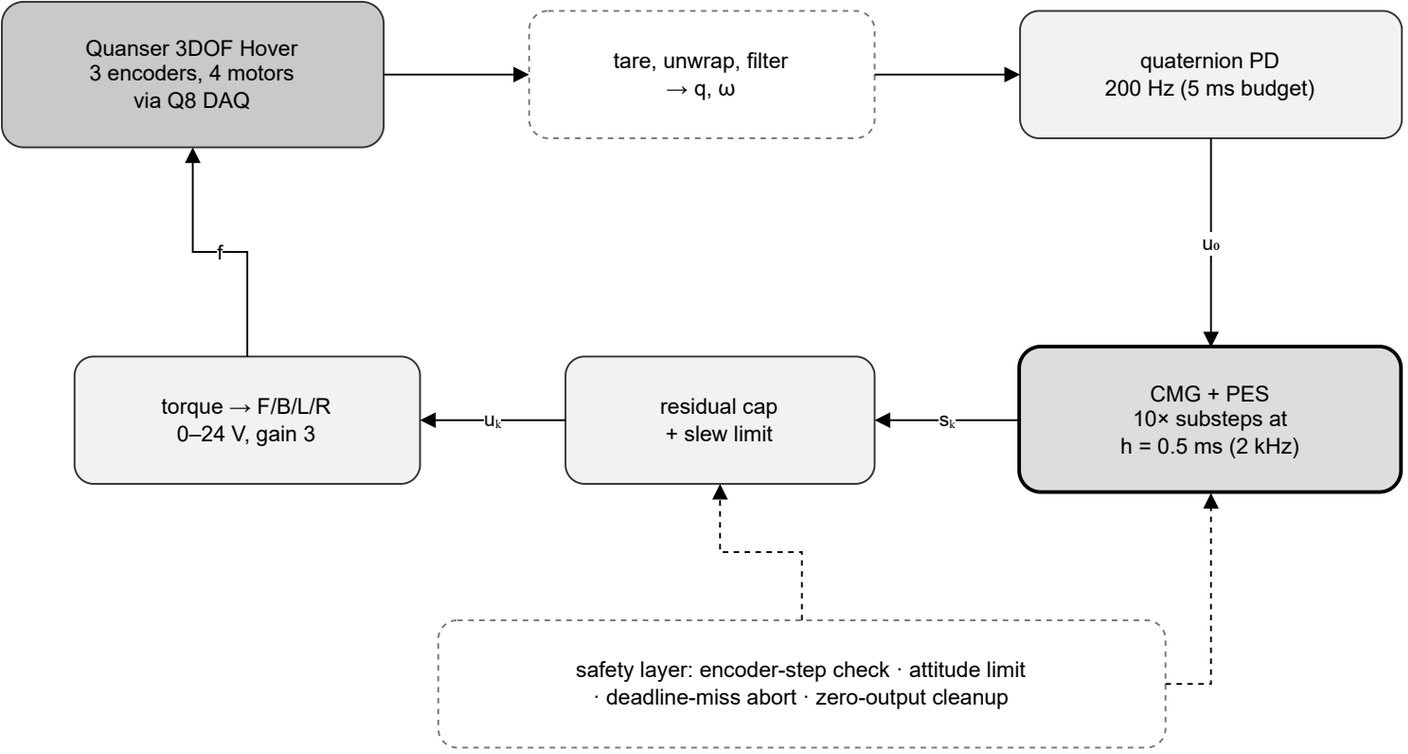
Each of the m command channels drives an antagonistic pair, so the population has 2m units. Activities are non-negative by construction; the sign is carried by the decoder. Initialising $W_0 = [I_m, -I_m]$ makes the untrained residual exactly the pair difference, so learning starts from a meaningful readout rather than a random one.



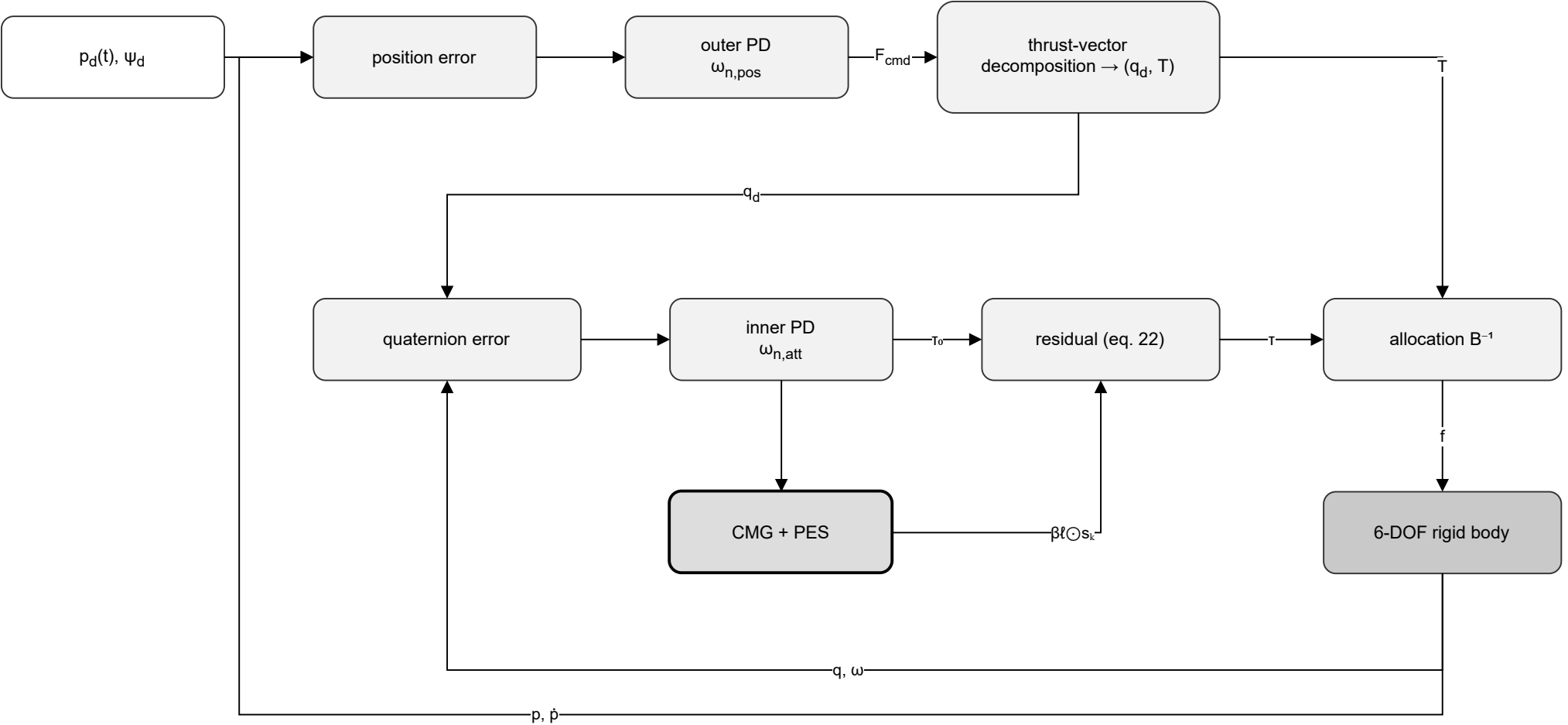
Both twins share reference, duration, integration step, actuator scale, allocator, initial condition, safety limits and logging. The residual path is the only difference, so the reported improvement is attributable to it rather than to re-tuning.



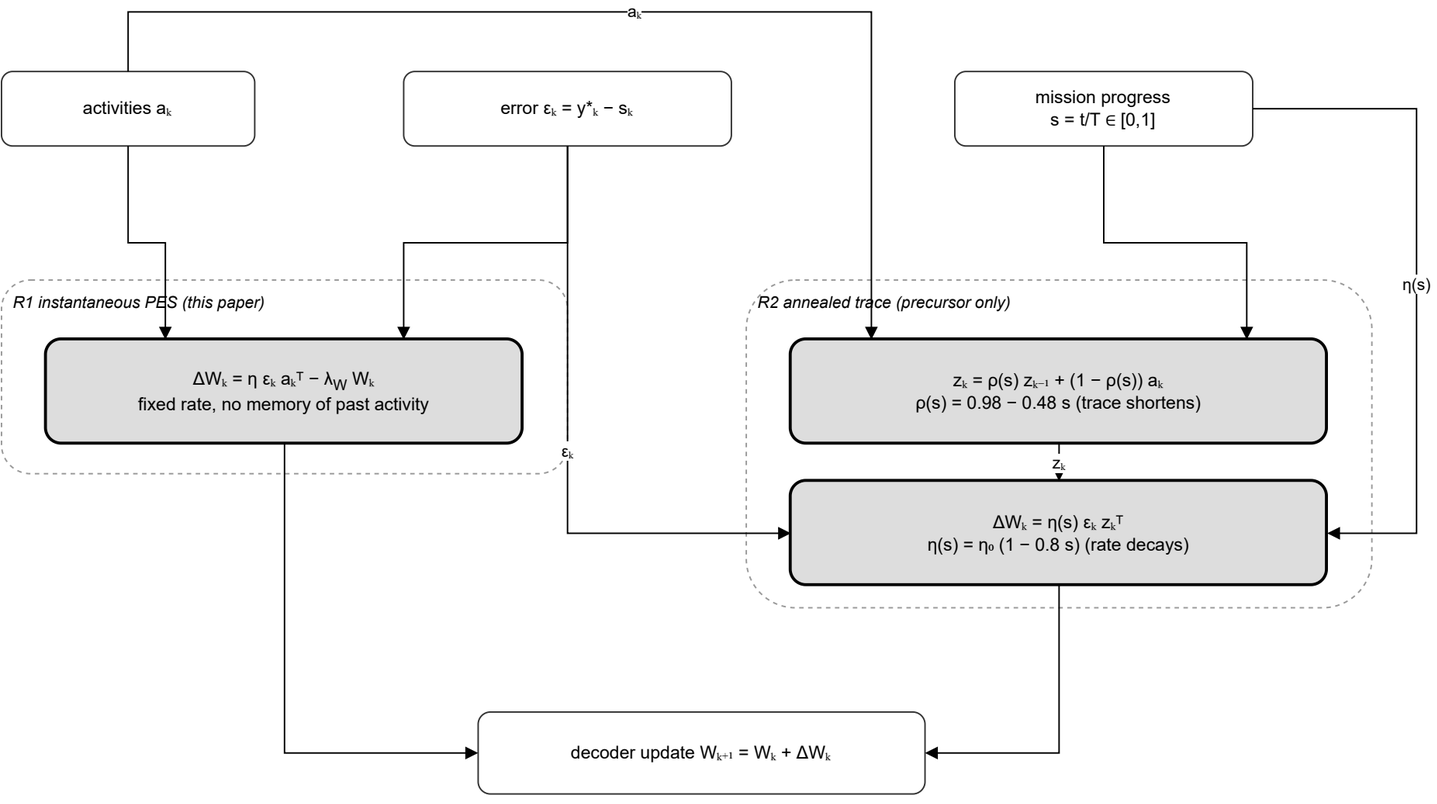
The CMG population is the model's only Ensemble: in Nengo a custom NeuronType can exist only inside one. Everything else is a Node, i.e. non-neural scaffolding. Current is injected at .neurons, bypassing the NEF encoders (gain = 1, bias = 0), and the PES learning rate is converted as $\eta_{nengo} = \eta\ (2m) / h$ so the direct and Nengo realizations coincide.



Hard real time: each 200 Hz hardware step contains ten 0.5 ms neural substeps, so the neural state advances at the same rate as in simulation. The controller is driven directly from Nengo; the same model under the Nengo GUI produces interpreter-lock contention and deadline-miss aborts.



Time-scale separation is required, not optional: the validated ceiling is $\omega_{n,pos}$ = [1.0, 1.0, 1.2] against $\omega_{n,att}$ = [4, 4, 3.5] rad/s. Both q_d and the measured q must be expressed in the same quaternion convention before the error is formed.



Both rules share the same activities, the same error signal and the same projected decoder; they differ only in what multiplies the error and how fast it is applied. R2 explores early and consolidates late. Its outcome is mission dependent — beneficial in some missions, adverse in others — so it is reported separately rather than folded into the main results.